Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
8	(a)	(i)	I.	Energy of photon	1			1		
			II.	[Minimum] energy needed to release an electron from a {surface / material / metal} Don't accept electron**s**	1			1		
		(ii)		**For barium** $E_{\text{k max}} = (6.63 \times 10^{-34} \times 7.9 \times 10^{14} - 4.03 \times 10^{-19})$[J] [= 1.21×10^{-19} J] **or** $E > \phi$ with $E = 5.2 \times 10^{-19}$ [J] (1) $V_{\text{stop}} = 0.76\text{ V}$ **ecf** (1) **For magnesium** Emission shown not to be possible i.e. $E < \phi$ **or** if $E_{\text{k max}} =$ - 6.2×10^{-20} [J] shown, needs a comment (1) If no other mark gained, give one mark for clear statement that photon energy is 5.2×10^{-19} [J] **Alternative for 1st and 3rd mark using threshold frequency.** Using: $hf_0 = \phi$, f_0 for barium = 6.08×10^{14} [Hz] and $7.9 \times 10^{14} > 6.08 \times 10^{14}$ [so possible] (1) f_0 for magnesium = 8.83×10^{14} [Hz] and $7.9 \times 10^{14} < 8.83 \times 10^{14}$ [so not possible] (1)		3		3	2	

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
	(b)			**Any 3 × (1) from:** • No greenhouse gases emitted [during operation] / decrease use of non-renewables / less demand on National Grid • Cost **or** insurance considerations • May not be pleasing to the eye (especially on old buildings) • Making the panels may damage environment or equiv • Some roofs face North and may not produce much power or may not be strong enough to support the panels • Night-time or heavy cloud makes available energy intermittent • Creation of jobs • Unethical to force people Don't accept photovoltaic panels are renewable or sustainable or eco-friendly			3	3		
				Question 8 total	**2**	**3**	**3**	**8**	**2**	**0**